navbea

Navbea RTLS Solutions Catalogue

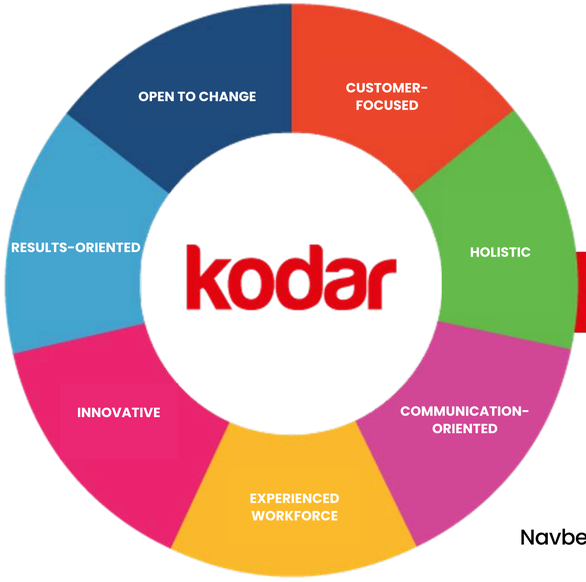
Real-Time Location Solutions

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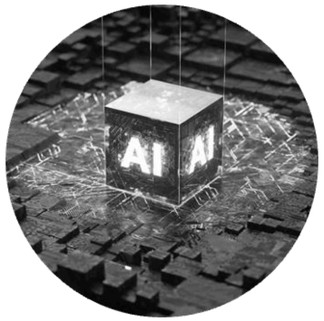




ABD Export BLE Sensor Production



Machine Monitoring and Remote Management Solutions



AI Computer Vision Deep Learning Has Started



Artificial Intelligence Occupational Safety Solutions Have Been Implemented



We have started working on international projects.

2014 2015 2016 2017 2018 2019 2020 2021 2022 2023

Navbea R&D Project

Technological Product Experience Certificate for the Development of IoT-Based Efficiency and Occupational Health and Safety Applications

NASA Space Center Houston Solution Provider

Implementation of RTLS Systems in 4 Major Theme Parks in Azerbaijan and 4 Major Thermal Power Plants

The European office based in Amsterdam, Netherlands has been opened.













Internet of Things (IoT) and Artificial Intelligence for

Real-Time Location Solutions (RTLS)

Navbea is an IoT (Internet of Things) and RTLS (Real-Time Location) platform that offers the ability to monitor and control personnel, equipment, and sensor data through various electronic devices and software products.

The Role of IoT and Location Intelligence in Enterprise Digitization



As enterprises evolve toward smart and data-driven operations, location intelligence plays a critical role in enabling visibility, automation, and compliance. Navbea supports this transformation by providing a scalable RTLS platform tailored to the needs of various industries.

From manufacturing safety to healthcare compliance, Navbea's systems deliver value by integrating seamlessly into corporate digital infrastructure without disrupting operations.

Navbea's **Location Technologies**

Navbea integrates multiple positioning technologies to ensure flexible, accurate, and cost-effective deployment.

BLE (Bluetooth Low Energy)

Low-cost, long-battery wearable tags and sensors.

UWB (Ultra-Wideband)

High-precision indoor tracking with 10–30 cm accuracy.

RFID

Passive and active tag tracking for asset and inventory monitoring.

GPS

Outdoor and vehicle tracking for mobile assets.

LoRa

Long-range low-power wireless for large-scale industrial coverage.

WiFi Integration

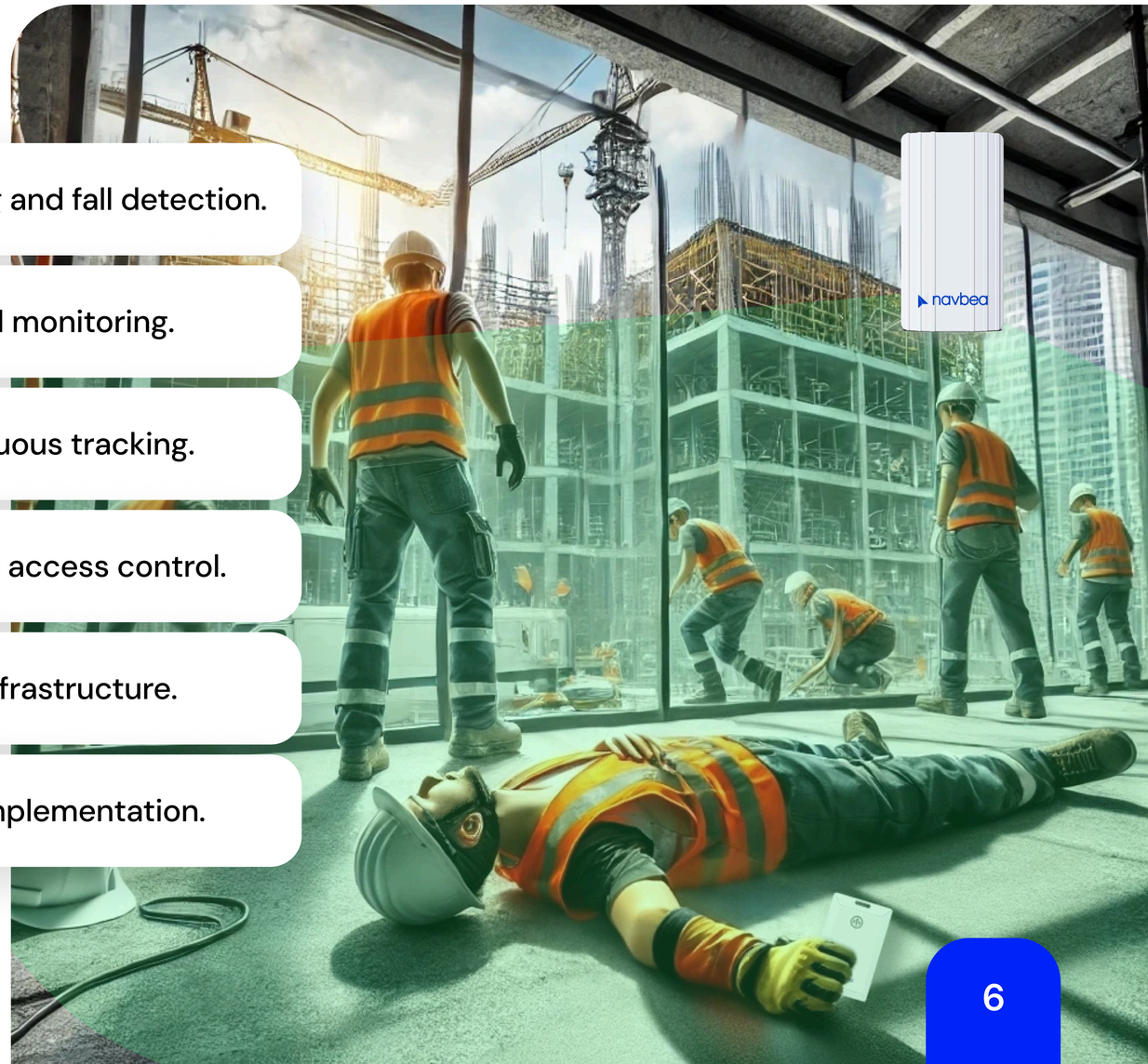
Uses existing access points from vendors like Cisco, HP, Huawei as RTLS gateways to reduce hardware investments.

AI-Enhanced Video Analytics

Optional integration with IP cameras for visual verification and behavior detection.

Benefits Delivered by Navbea RTLS Solutions

- Improved workforce safety through real-time emergency tracking and fall detection.
- Operational efficiency via task automation and zone-based monitoring.
- Asset and equipment utilization optimization through continuous tracking.
- Audit and compliance readiness via historical location logs and access control.
- Infrastructure cost reduction by utilizing existing wireless infrastructure.
- Faster ROI through modular deployment and sector-based implementation.



Sectors & Use Cases



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- Real-Time Subcontractor Tracking
- Emergency Assembly Point Automation
- Restricted Zone Access Control

02

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- Student Entry/Exit and Attendance Automation
- Campus Zone Access and Area Safety
- Real-Time Parental Notification Integration

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- Medical Equipment Tracking and Loss Prevention
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- Emergency Response via Panic Button

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- Lost Child Tracking and Parent Notification
- Visitor Flow and Queue Optimization
- Equipment Monitoring and Fault Response

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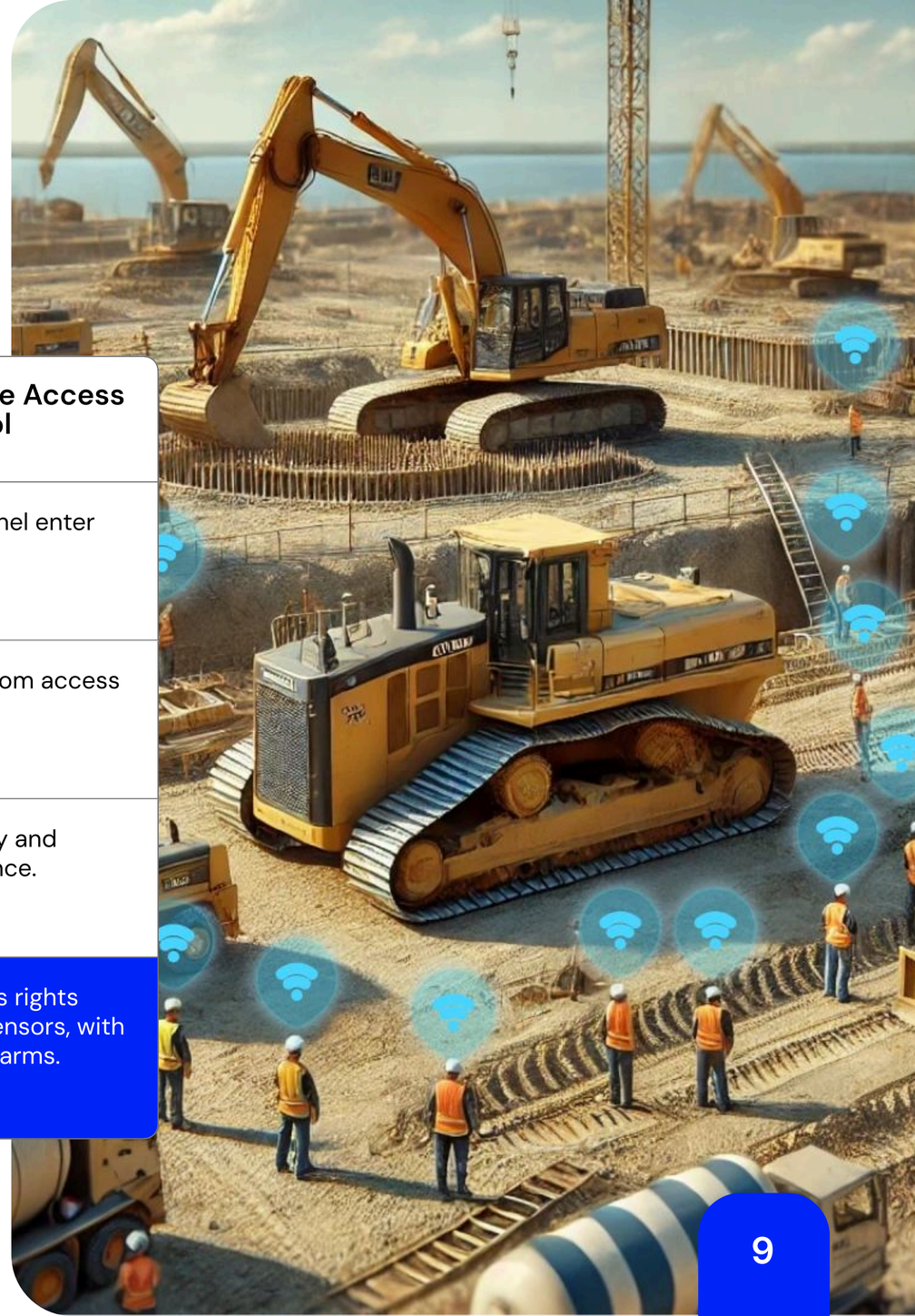
GOVERNMENT INSTITUTIONS

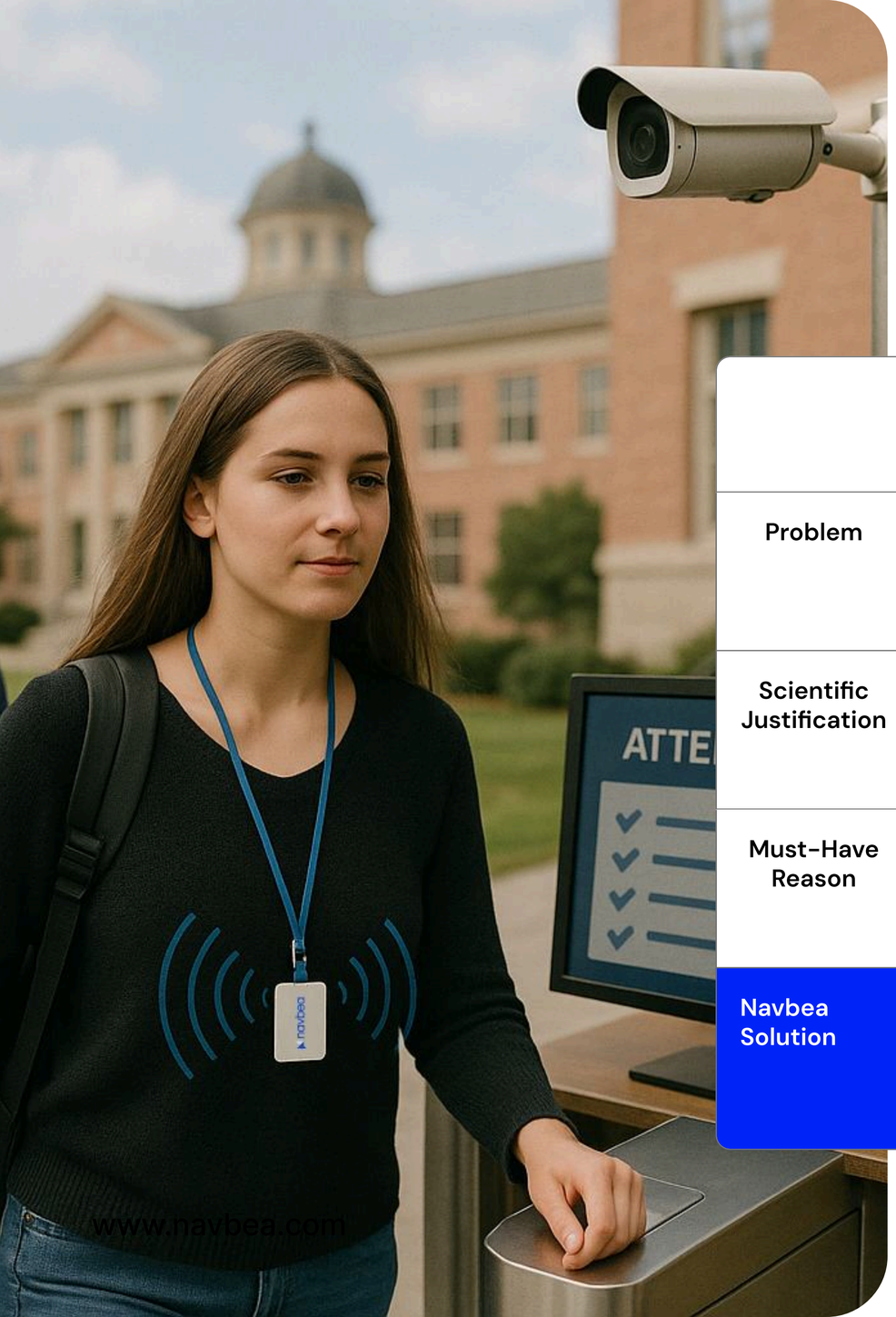
- Visitor Tracking and Access Monitoring
- Staff Attendance and Task Allocation
- Emergency Evacuation and Muster Point Automation



CONSTRUCTION

	Real-Time Subcontractor Tracking	Emergency Assembly Point Automation	Restricted Zone Access Control
Problem	Uncontrolled movement and unauthorized access by subcontractors on-site.	Manual headcounts are inaccurate during emergency evacuations.	Unqualified personnel enter hazardous areas.
Scientific Justification	25% higher accident rate on sites without tracking (EU-OSHA).	Up to 30% error in drills (British Safety Council).	70% of falls stem from access breaches.
Must-Have Reason	Legal compliance and audit readiness.	EU emergency planning mandates real-time counts.	Required for liability and insurance compliance.
Navbea Solution	Wearable BLE/LoRa badges track entry-exit and assigned work zones in real time.	BLE-powered automatic headcount and live dashboard alerting in designated assembly zones.	Zone-based access rights enforced via BLE sensors, with real-time breach alarms.





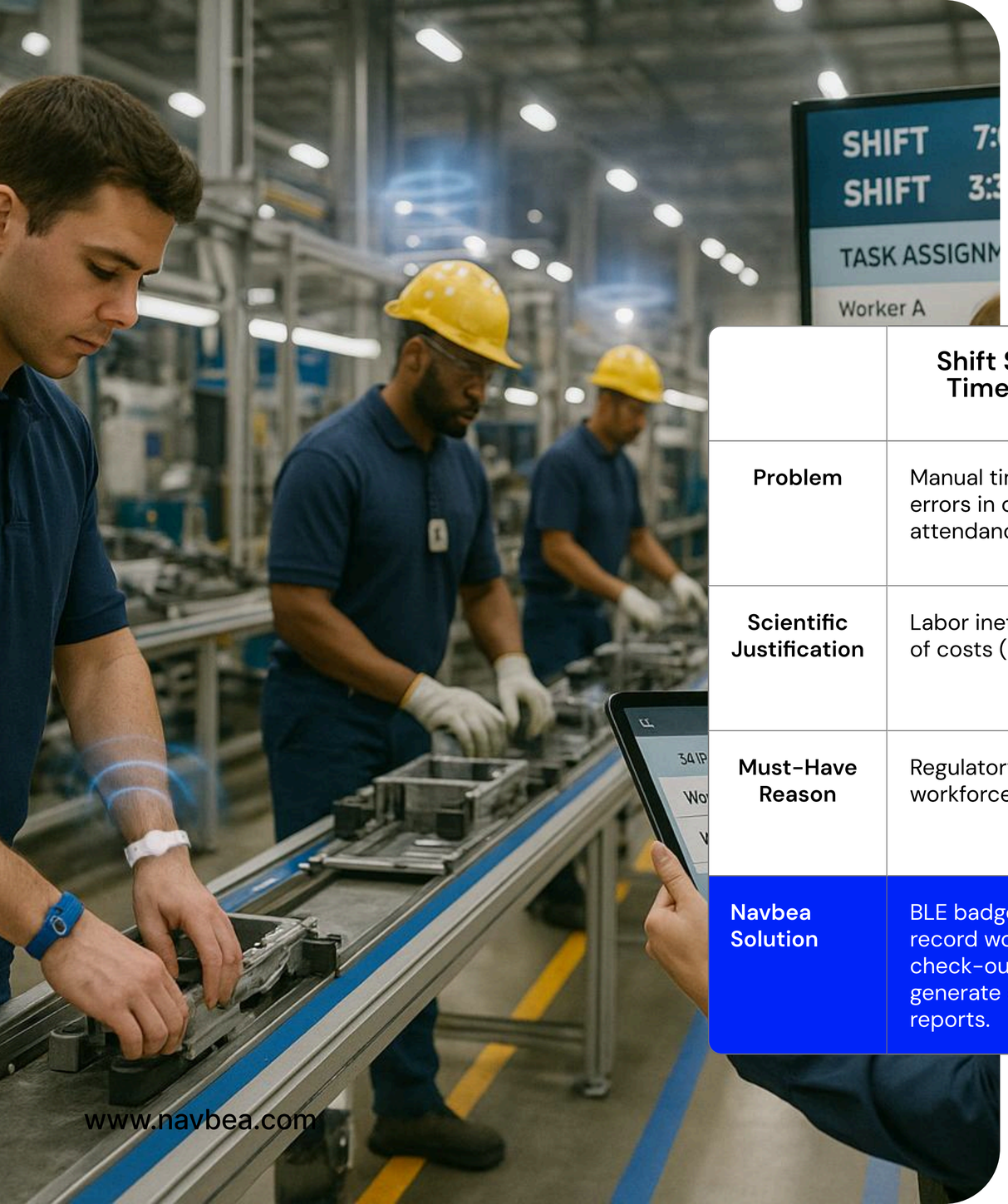
EDUCATION

	Student Entry/Exit and Attendance Automation	Campus Zone Access and Area Safety	Real-Time Parental Notification Integration
Problem	Attendance is manually recorded and school exits go unmonitored.	Students can enter unauthorized or hazardous areas without oversight.	Parents lack live visibility of their child's presence on campus.
Scientific Justification	18% absentee reduction (UNESCO).	45% of incidents involve zone violations.	Higher satisfaction via transparency.
Must-Have Reason	Guardian reporting requirements.	EU campus safety protocols.	Family engagement laws promote it.
Navbea Solution	BLE-enabled student ID cards provide automated attendance logs and track school entries and exits.	Area-based permissions and BLE geofencing alert staff when students enter restricted zones.	BLE data is shared via secure mobile apps, notifying parents of entry, exit, and current location in real time.

HEALTHCARE

	Medical Equipment Tracking and Loss Prevention	Patient Safety and Area Access Control	Emergency Response via Panic Button
Problem	Devices are frequently misplaced, underutilized, or used in the wrong location.	Patients unintentionally enter restricted or unsafe zones.	Patients or staff have no direct way to summon help during medical emergencies.
Scientific Justification	15% misplacement rate (EU).	60% higher risk in unsafe areas.	40% higher survival with <2 min response (Lancet).
Must-Have Reason	MDR compliance.	Safety protocol enforcement.	Mandatory for high-risk environments.
Navbea Solution	BLE-tagged medical equipment enables real-time tracking, movement history, and theft alerts.	BLE patient wristbands provide real-time tracking and alert staff upon entry to unauthorized zones.	BLE badges or wearables with built-in panic buttons trigger immediate alerts to on-duty staff.



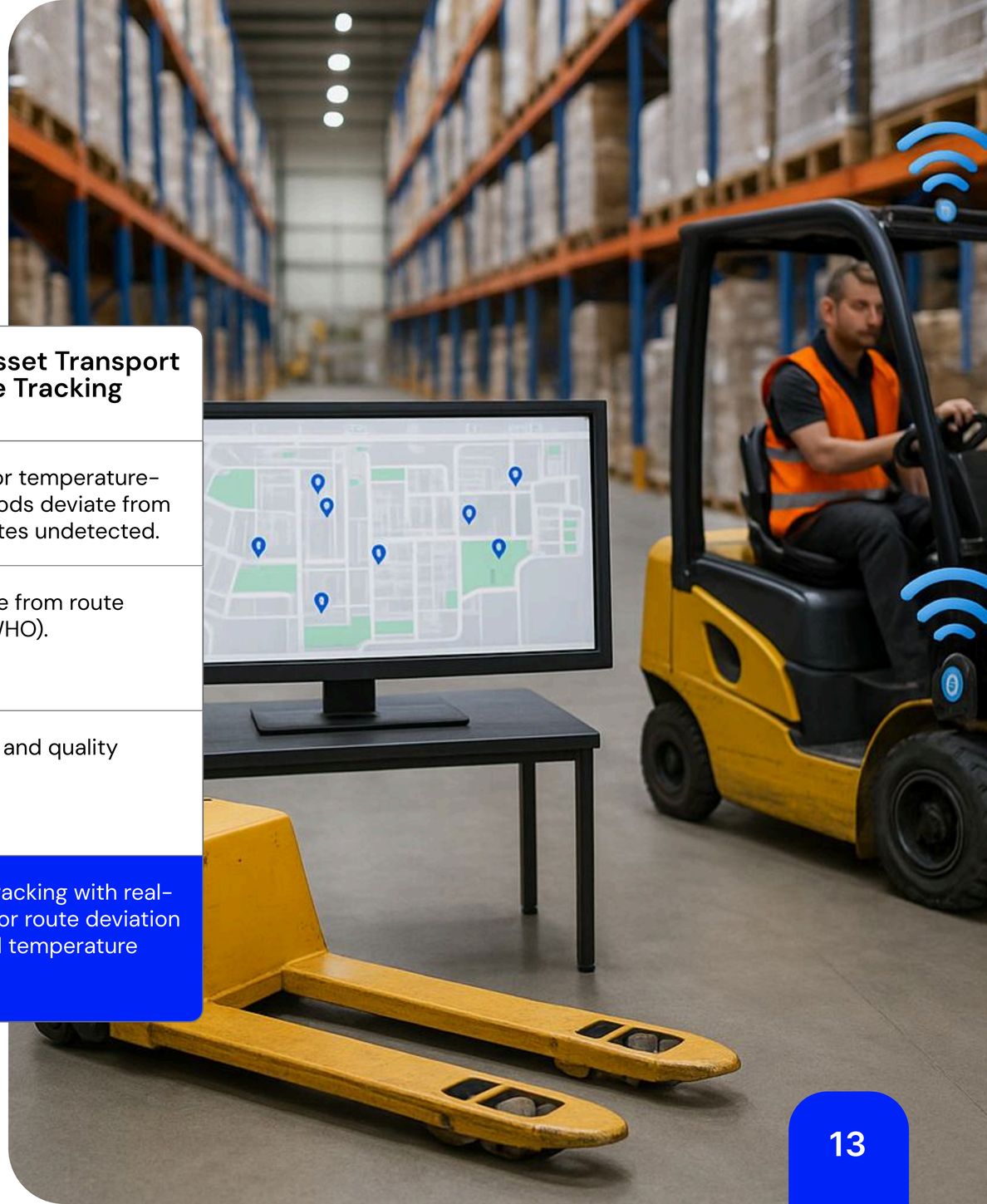


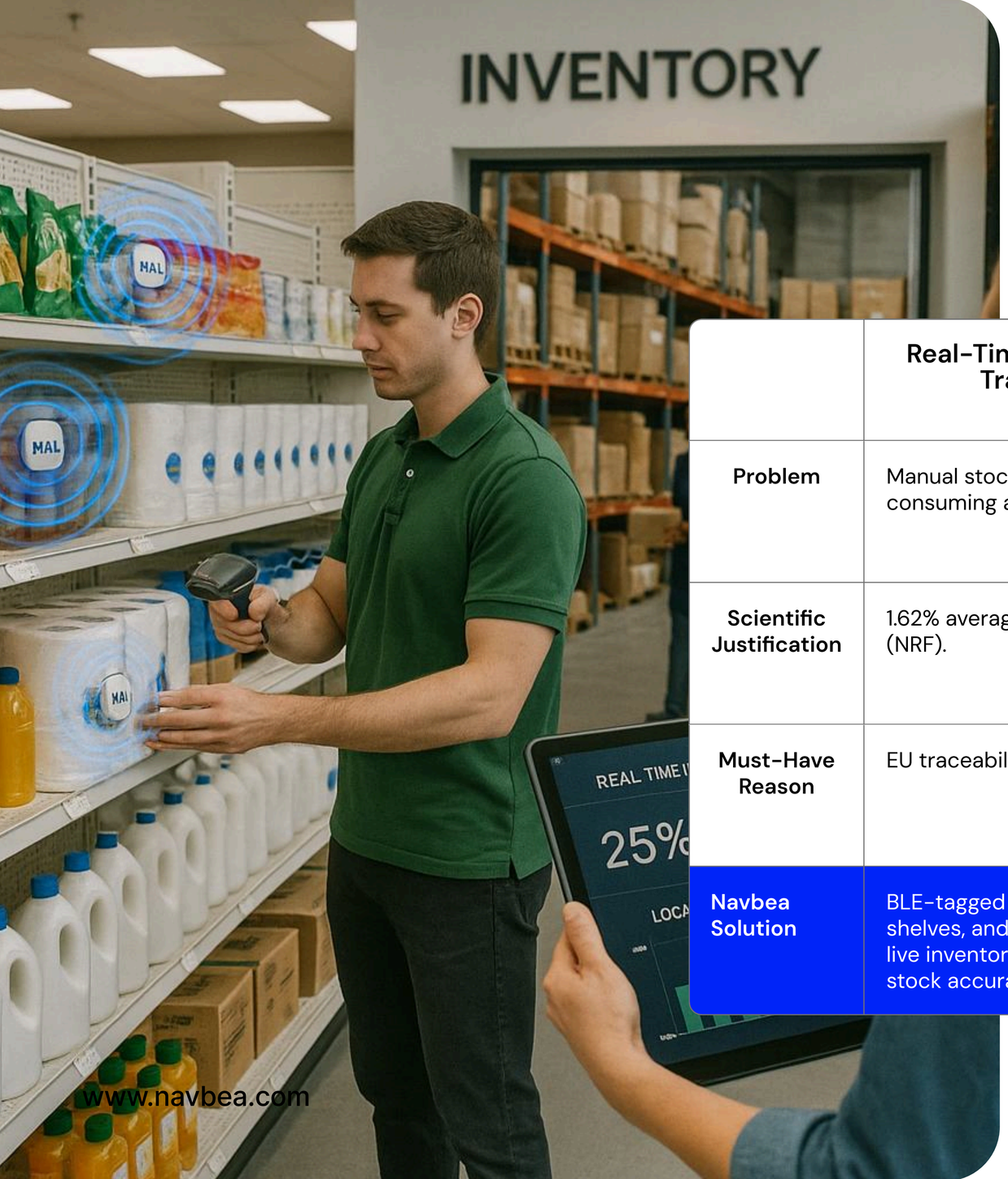
MANUFACTURING

	Shift Start and End Time Automation	Fall and Inactivity Detection	Equipment Utilization and Efficiency Monitoring
Problem	Manual timekeeping leads to errors in overtime and attendance tracking.	Injuries or accidents in isolated areas go unnoticed, delaying response.	No insight into how often, where, or by whom machinery is being used.
Scientific Justification	Labor inefficiencies = 20% of costs (McKinsey).	>6 min delay increases fatality risk 50% (WHO).	15% productivity loss (Deloitte).
Must-Have Reason	Regulatory reporting and workforce analytics.	Real-time alerting required for HSE compliance.	Predictive maintenance needs usage data.
Navbea Solution	BLE badges automatically record worker check-in/ check-out times and generate compliance-ready reports.	Wearable BLE sensors detect falls and prolonged inactivity, triggering real-time alerts.	BLE-tagged assets enable real-time monitoring of location, usage time, and operator behavior.

LOGISTICS

	Forklift Tracking and Safe Navigation	Subcontractor Entry/ Exit Monitoring	Critical Asset Transport Route Tracking
Problem	Forklifts operate without route monitoring, leading to accidents and misuse.	External workers move throughout the site without digital visibility.	High-value or temperature-sensitive goods deviate from planned routes undetected.
Scientific Justification	38% of incidents due to routing issues (OSHA).	30% annual rise in security incidents (TAPA).	25% spoilage from route deviation (WHO).
Must-Have Reason	Required for safety compliance.	Legal accountability.	Compliance and quality control.
Navbea Solution	BLE-based tracking provides location, speed, and zone control for each vehicle.	BLE badges log subcontractor presence, time spent, and zone-level access in real time.	BLE + GPS tracking with real-time alerts for route deviation and optional temperature logging.



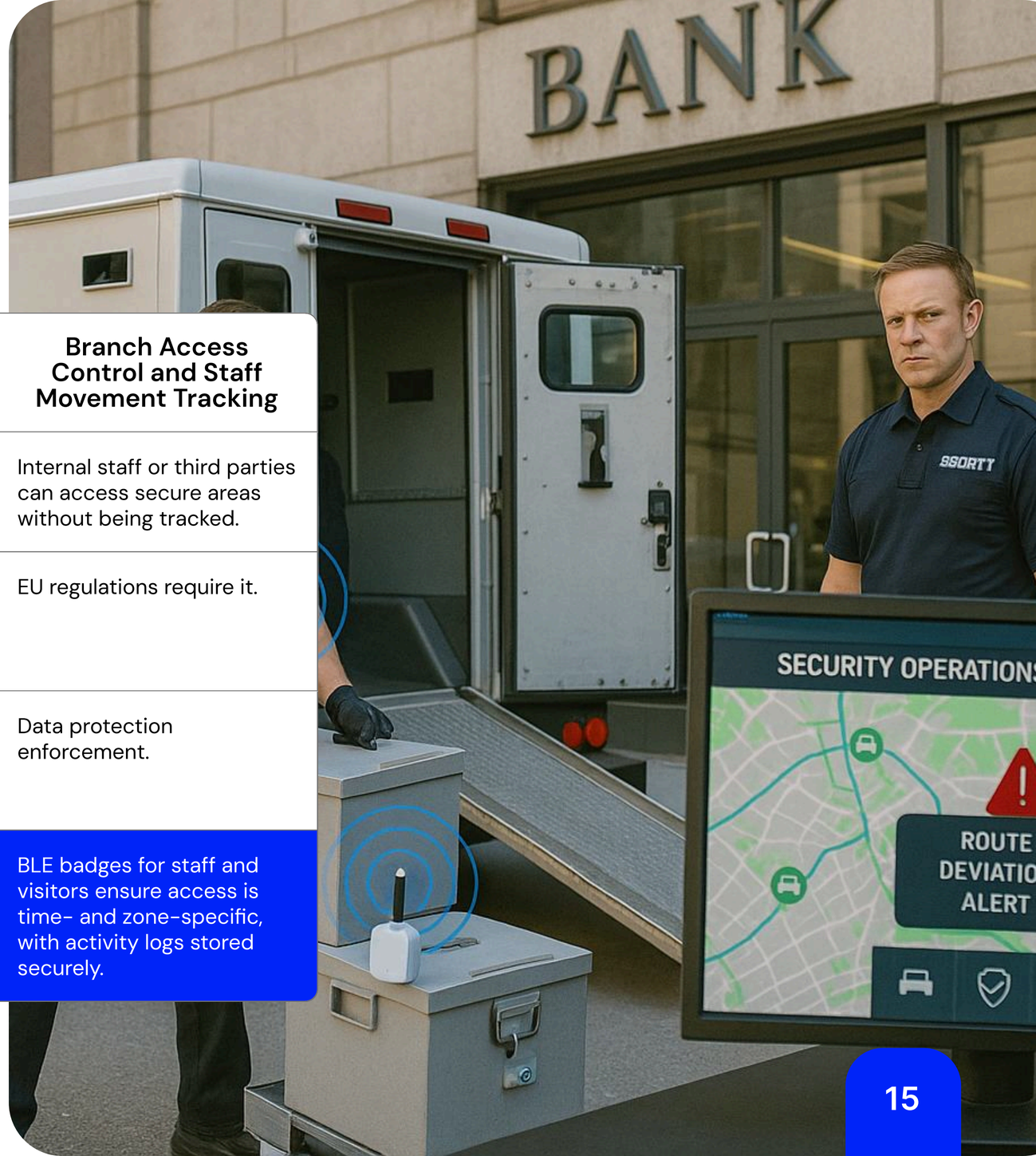


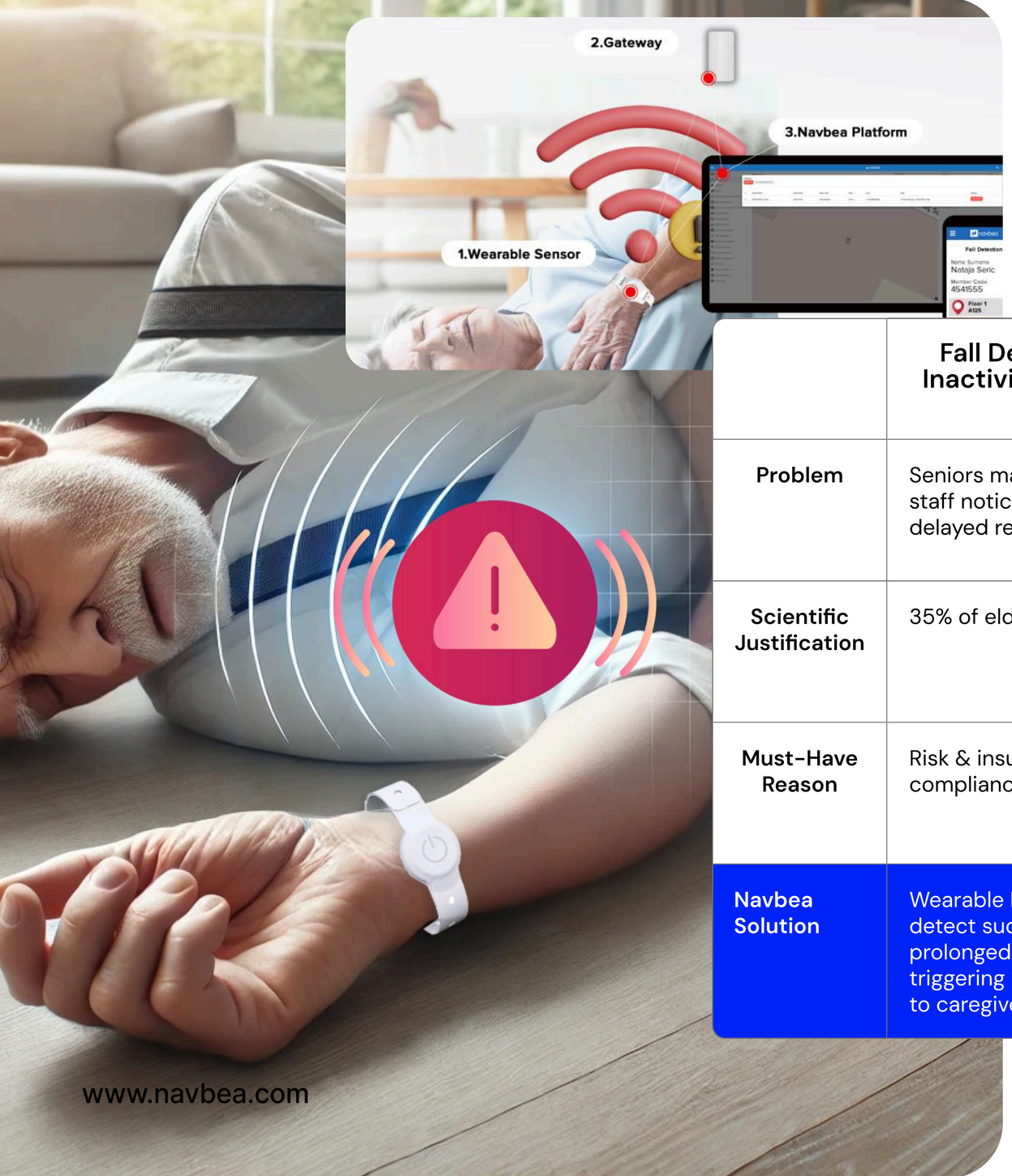
RETAIL

	Real-Time Inventory Tracking	Theft Prevention and Loss Detection	Automated Replenishment and Shelf Monitoring
Problem	Manual stocktaking is time-consuming and inaccurate.	Internal shrinkage and theft go undetected without real-time monitoring.	Empty shelves are restocked late due to manual inspection.
Scientific Justification	1.62% average shrinkage (NRF).	35% shrinkage due to internal losses.	Up to 8% loss (McKinsey).
Must-Have Reason	EU traceability mandates.	Insurance and audit proof.	Efficient supply chain operation.
Navbea Solution	BLE-tagged merchandise, shelves, and assets enable live inventory tracking and stock accuracy over 98%.	BLE sensors trigger movement alarms and retain historical tracking data for stolen or misplaced items.	BLE-based shelf tagging and integration with ERP systems automate low-stock alerts and reorder triggers.

FINANCE

	Cash Transport Monitoring and Security	IT Asset and Equipment Inventory (ATMs, Servers, POS Devices)	Branch Access Control and Staff Movement Tracking
Problem	Cash boxes can go off-route or be stolen without real-time alerts.	IT equipment locations are often unknown, causing delays in maintenance and risk of loss.	Internal staff or third parties can access secure areas without being tracked.
Scientific Justification	72% robberies target transport (EU).	28% record errors (KPMG).	EU regulations require it.
Must-Have Reason	PSD2/AML requires traceability.	ISO 27001/GDPR mandate audit trails.	Data protection enforcement.
Navbea Solution	BLE + GPS sensors track cash containers and vehicles in real time with deviation alerts and audit trails.	BLE-tagged equipment enables real-time location tracking, usage history, and scheduled maintenance logging.	BLE badges for staff and visitors ensure access is time- and zone-specific, with activity logs stored securely.



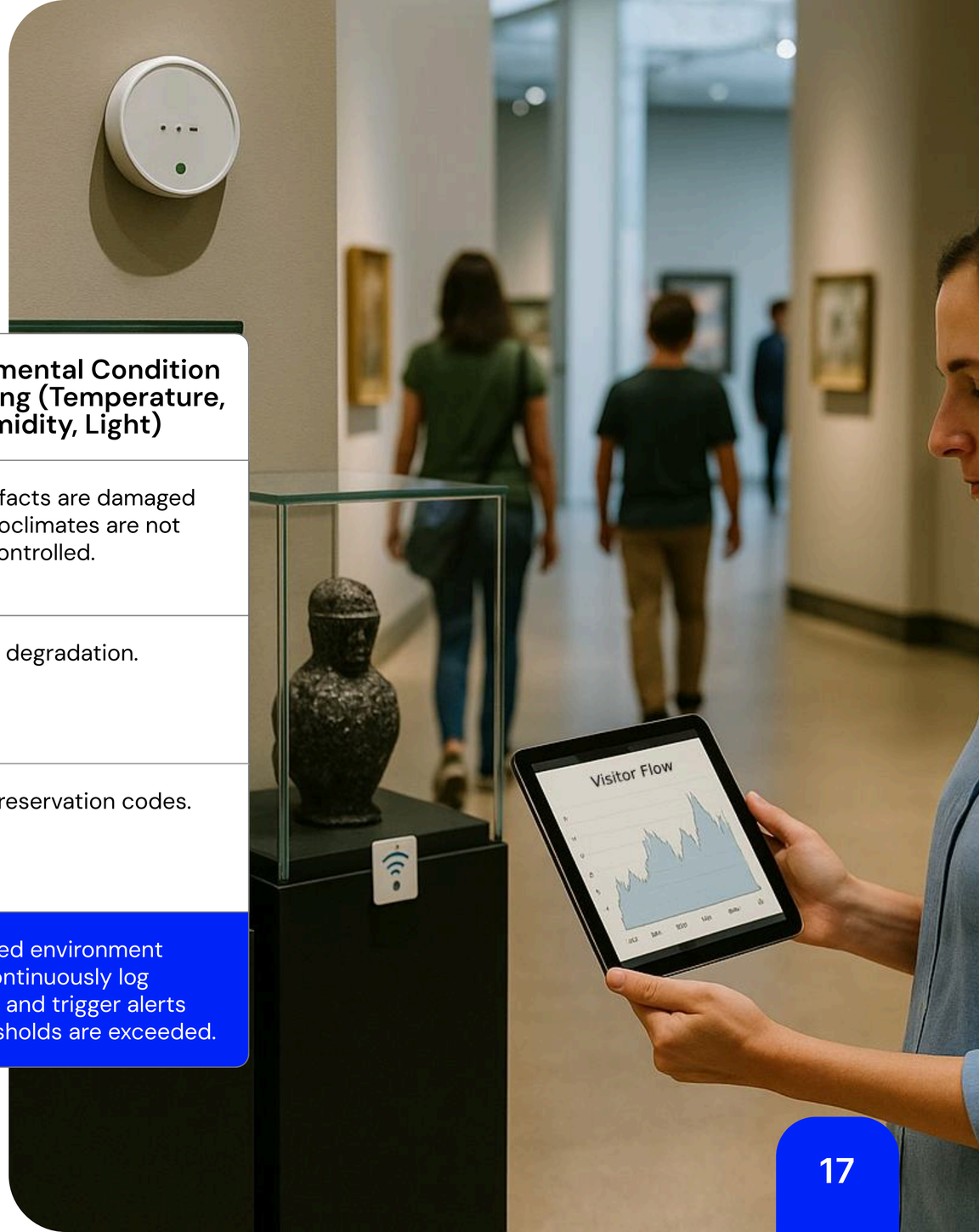


ELDERLY CARE

	Fall Detection and Inactivity Monitoring	Bed Exit Monitoring and Night Movement Detection	Visitor Monitoring and Contact Duration Tracking
Problem	Seniors may fall without staff noticing, leading to delayed response.	Seniors, especially with dementia, may leave beds at night unnoticed.	Unregulated visitor activity increases exposure risk and reduces care quality.
Scientific Justification	35% of elderly fall yearly.	60% of Alzheimer's patients affected.	Post-pandemic care rules demand this.
Must-Have Reason	Risk & insurance compliance.	Safety compliance.	Infection and safety standards.
Navbea Solution	Wearable BLE wristbands detect sudden falls and prolonged inactivity, triggering immediate alerts to caregivers.	BLE-enabled motion and doorframe sensors detect bed exits and monitor nighttime movement patterns.	BLE badges for visitors track duration, movement, and contact density with sensitive zones.

MUSEUMS

	Artifact Security and Anti-Theft Monitoring	Visitor Flow and Density Management	Environmental Condition Monitoring (Temperature, Humidity, Light)
Problem	Portable or small artifacts can be stolen or misplaced without immediate detection.	Congestion in popular exhibit areas affects safety and visitor satisfaction.	Fragile artifacts are damaged when microclimates are not properly controlled.
Scientific Justification	7% loss rate (Interpol).	40% longer evacuation time.	25% faster degradation.
Must-Have Reason	Directive 93/7/EEC.	Venue capacity laws.	Heritage preservation codes.
Navbea Solution	BLE + RFID tags provide real-time location tracking, movement alerts, and tamper detection.	Area-based BLE sensors generate live heatmaps, guide flows via digital signage, and trigger crowding alerts.	BLE-enabled environment sensors continuously log conditions and trigger alerts when thresholds are exceeded.



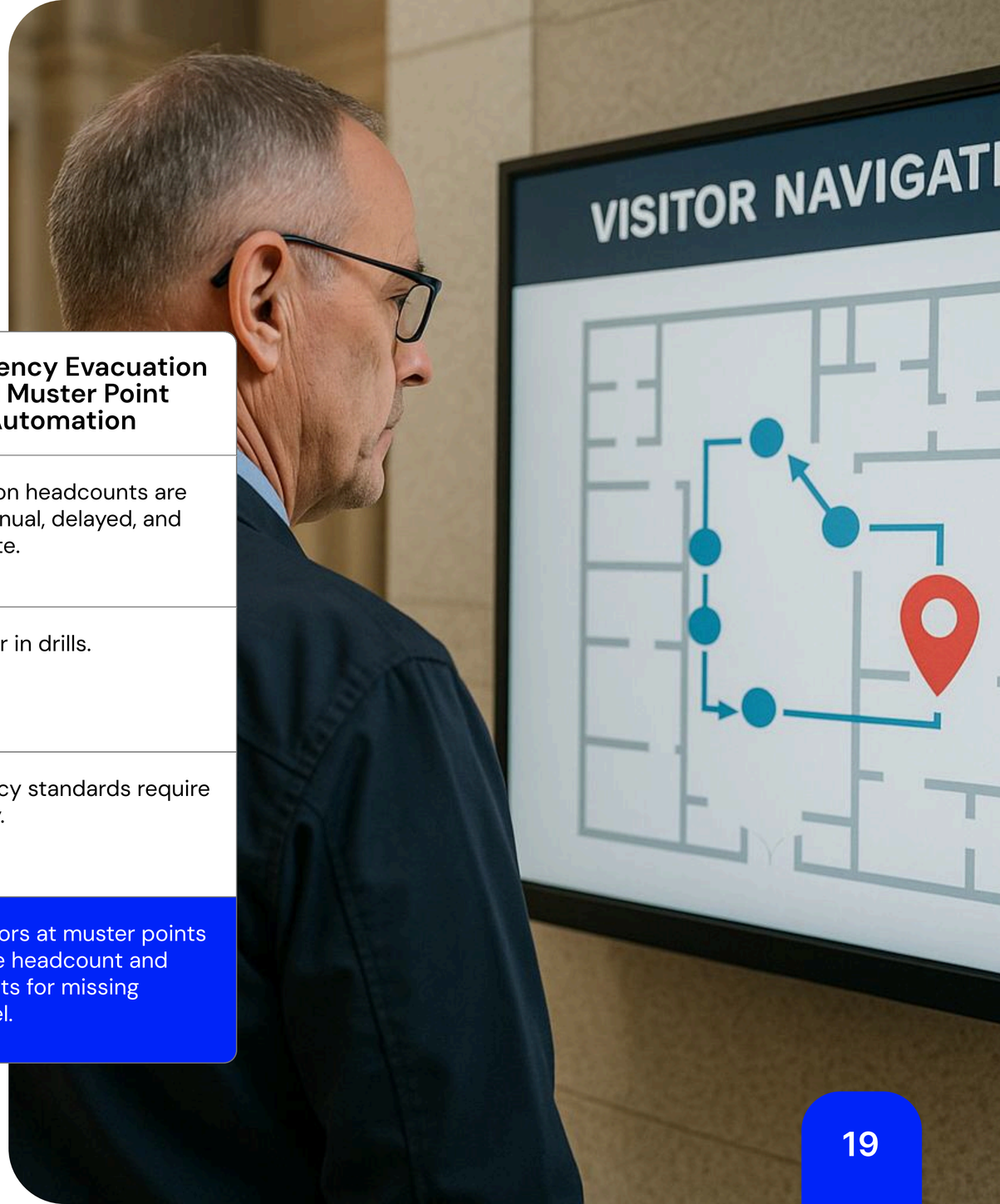


THEME PARKS & ENTERTAINMENT

	Lost Child Tracking and Parent Notification	Visitor Flow and Queue Optimization	Equipment Monitoring and Fault Response
Problem	Children can easily get separated from guardians in large, crowded environments.	Long queues and overcrowding reduce guest satisfaction and operational efficiency.	Ride malfunctions or maintenance delays go unnoticed, increasing safety risk.
Scientific Justification	5,000+ cases/year (IAAPA).	40% lower revisit rates.	60% more risk with late action.
Must-Have Reason	Licensing & safety mandates.	Guest flow efficiency.	EU ride safety law.
Navbea Solution	Children wear BLE wristbands that provide real-time location data, with automated mobile alerts sent to parents.	BLE sensors track visitor density, optimize queue routing, and integrate with signage for live guidance.	BLE-tagged assets enable real-time performance tracking, fault alerting, and technician dispatch logs.

GOVERNMENT INSTITUTIONS

	Visitor Tracking and Access Monitoring	Staff Attendance and Task Allocation	Emergency Evacuation and Muster Point Automation
Problem	Government facilities lack real-time insight into where visitors go after entry.	Staff activity and productivity across departments are not digitally monitored.	Evacuation headcounts are often manual, delayed, and inaccurate.
Scientific Justification	€1.2B annual risk (EU).	5% inefficiency loss (OECD).	30% error in drills.
Must-Have Reason	Institutional access rules.	Internal audit requirements.	Emergency standards require accuracy.
Navbea Solution	BLE-enabled badges provide real-time zone tracking and entry-point analytics.	BLE badges track time spent per task and area, generating reports for accountability and optimization.	BLE sensors at muster points automate headcount and send alerts for missing personnel.



Customers

Human Resources

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YENİKÖY KEMERKÖY
Yerli Kaynak Güçlü Enerji

Aydem enerji

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